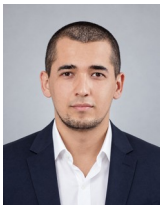
Sean Kakhkhorov



Full-Stack Software Engineer (TypeScript · Next.js · NestJS · FastAPI)

Suwon, South Korea | kakhkhorov97@gmail.com | +82-10-7901-7555
Portfolio: https://www.sean-kakhkhorov.site | GitHub: https://github.com/kakhkhorov97
Languages: Korean (conversational), English (fluent)

PROFESSIONAL SUMMARY

Full-stack software engineer with 2 years 10 months of professional experience building and deploying production web applications, REST/GraphQL APIs, and Linux infrastructure for 20+ client companies, plus 4 end-to-end AI web platforms from data modeling to cloud deployment. Experienced in transactional concurrency control, Toss Payments verification, hybrid RAG search, and CI/CD. Conversational Korean and fluent English.

TECHNICAL SKILLS

Languages and Frontend: TypeScript, JavaScript, Python, SQL, React, Next.js (App Router), Tailwind CSS, Apollo Client, Redux Toolkit, next-intl

Backend and Data: Node.js, NestJS, Express, FastAPI, GraphQL, REST, Socket.IO, JWT, OAuth 2.0 (PKCE), PostgreSQL, Drizzle ORM, MongoDB, Redis, BullMQ

Applied AI and RAG: Gemini API (structured output, embeddings), hybrid RAG (BM25 + vector, RRF), pgvector, ChromaDB, Kiwi Korean tokenizer, Zod, Pydantic

Infrastructure and QA: Docker, Ubuntu Linux, Nginx, Google Cloud Run, GitHub Actions (CI/CD), Cloudflare, Jest, Vitest, Supertest, pytest (700+ tests)

PROFESSIONAL EXPERIENCE

IT Local — Full-Stack Web Developer | Tashkent, Uzbekistan (B2B web and software agency) | Nov 2023 – Feb 2026

- Built and maintained production web applications and APIs for 20+ client companies as part of a core team of 4 developers covering frontend, backend, database, and deployment (React, Node.js, authentication, input validation, index-backed schema design).
- Built websites and data-collection web systems for the agency's largest client, a construction company building houses and apartments across Uzbekistan, covering construction progress, materials and inventory, apartment sales and bookings, and site reports, used by customers and office staff.
- Provisioned, secured, and maintained 5–15 live Ubuntu servers (Nginx, Docker, PM2, Cloudflare); replaced manual SSH deploys with GitHub Actions CI, cutting releases to about 4–5 minutes.

Atlantis Investment — Backend Developer | Tashkent, Uzbekistan (remote, alongside studies) | Apr 2023 – Sep 2023

- Built Express.js REST APIs with JWT/session authentication and file uploads; designed compound MongoDB indexes for search and pagination.

PROJECTS

Sofiya — AI Book Concierge and E-Commerce Store | Next.js 15, NestJS 11, PostgreSQL (pgvector), Redis, Cloud Run
Live: https://sofiya.store/ | **Code:** https://github.com/kakhkhorov97/Sofiya-Bookstore/tree/v2

- EN/KO/RU concierge fusing pgvector and full-text search via RRF with in-stock filtering and verified citations: 100% Hit@5, 93% Hit@1, 0.95 MRR (vs. 85% Hit@5 for keyword search).
- Toss Payments with server-side amount validation, idempotent checkout, and automatic refunds; keyless OIDC CI/CD to Cloud Run, Google OAuth (PKCE), Redis rate limiting.

CreatorHub — Multilingual Creator Marketplace | Next.js, NestJS (GraphQL), MongoDB, Socket.IO, Docker, Nginx | Jul 2026 – Present
Live: https://creatorhub.site/ | **Code:** https://github.com/kakhkhorov97/CreatorHub

- EN/KO/UZ marketplace with Socket.IO chat, admin moderation, and a Gemini assistant; transaction-checked request keys and unique indexes prevent duplicate inquiries under concurrency.
- Restart-safe notification worker using 60-second distributed leases and batch checkpoints; Docker/Nginx VPS with GitHub Actions CI and health-checked automatic rollbacks.

Korea Visa Assistant — Bilingual Immigration Law RAG | FastAPI, Python, Kiwi, ChromaDB, BM25, Cloud Run
Live: https://korea-visa-rag-jwq77bpivq-uc.a.run.app/ | **Code:** https://github.com/kakhkhorov97/korea-visa-rag

- EN/KO RAG over 21 visa categories with statute chunking, Kiwi tokenization preserving visa codes, and BM25 + Chroma RRF: 96.9% Hit@5, 0.935 MRR (+33% over 0.702 BM25), with quote-checked citations, raw-statute fallback, and PII blocking.

CargoCheck AI — Trade Document Verification System | BullMQ, Redis, Gemini API, Zod, PostgreSQL, Docker
Live: https://cargocheck-ai-web.vercel.app/ | **Code:** https://github.com/kakhkhorov97/cargocheck-ai

- Cross-verifies invoices and packing lists with BullMQ workers, Zod-validated Gemini extraction, human-in-the-loop review, quantity/unit/decimal checks, PDF audit reports, and file signature checks.

EDUCATION AND LANGUAGES

Chonnam National University (Yeosu, South Korea) — B.A. in International Trade, taught entirely in Korean | Sep 2019 – Aug 2023

Languages: Korean (conversational), English (fluent), Uzbek (native), Russian (elementary)